

A few directions in optimal transport

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Matchings

Example

- ▶ distribution of buyers,
- ▶ distribution of sellers,
- ▶ how to match the buyers and sellers

Example

- ▶ pixels in one photo
- ▶ pixels in another photo
- ▶ how to match these pixels in different photos. (questions in image processing)

Example

- ▶ a distribution of cells in different cell types in an organism today
- ▶ a distribution of cells in different cell types in an organism tomorrow
- ▶ how the organism change from today to tomorrow.

Optimal transport as optimal matching

- ▶ Source and target domains X, Y , respectively.
- ▶ μ, ν source and target measures of *equal mass*, on X and Y , respectively.
 - ▶ μ is a distribution of $x \in X$, ν is a distribution of $y \in Y$.

A **transportation plan** π is a matching between μ and ν .

It is a measure on $X \times Y$, which satisfies

$$\int_Y d\pi(\cdot, y) = \mu(\cdot) \quad (\text{the first marginal of } \pi \text{ is } \mu),$$

$$\int_X d\pi(x, \cdot) = \nu(\cdot) \quad (\text{the second marginal of } \pi \text{ is } \nu),$$

$$\pi \geq 0.$$

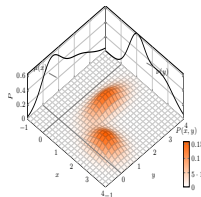


Image provided by Marco Cuturi

- ▶ In probabilistic terminology, a transport plan π is a **joint distribution** between μ and ν .

$\Pi(\mu, \nu)$ = the set of transport plans between μ and ν :

- ▶ **Transportation cost** $c(x, y)$ between x and y .
- ▶ Total transportation cost of a transport plan π :

$$\int_{X \times Y} c(x, y) d\pi(x, y).$$

- ▶ **The optimization problem:** Find the optimal solution $\bar{\pi}$ for

$$\inf_{\pi \in \Pi(\mu, \nu)} \int_{X \times Y} c(x, y) d\pi(x, y).$$

Brenier's theorem: characterization of optimal transport

- One of the most important theorems in optimal transport. It roughly says for the transportation cost function $c(x, y) = |x - y|^2/2$,

"An optimal transport plan is given by a mapping $T = \nabla \phi$ for a **convex** function ϕ ."

- For **more general type of cost** c , there is a generalization of this, called **Gangbo-McCann theorem**.

"Under appropriate conditions on c , the optimal transport plan is given by a mapping T , which is the **c -gradient** of a **c -convex** function."

- For example, $c(x, y) = \text{dist}(x, y)^p$, $p > 1$, $\text{dist}(x, y) = \text{Euclidean/Riemannian distance}$, etc. ...

Optimal transport is a simple concept and thus has connections to many things!

A few research directions (among many) where ideas of optimal transport are applied:

- ▶ **Connecting different distributions in an effective way**

- ▶ Developmental curves

- ▶ **Matchings under stochastic constraints**

- ▶ Matching probability measures in stochastic ways

- ▶ **Providing solutions to partial differential equations**

- ▶ Dynamics of phase transition/ free boundaries

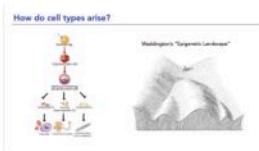
Connecting different distributions in an effective way:

Optimal transport can be used to connect (many) different distributions in an effective way, and can be applied to the problem of **inferring**

- ▶ developmental curves

from sample data.

Evolution of distributions, in the gene expression space



One can view the developmental process of cells as the **evolution of distributions**, $\mathbf{P}_t$, $0 \leq t \leq t_{max}$, on the gene expression space.

Figure courtesy of Geoffrey Schiebinger

Diverse cell types arise during reprogramming

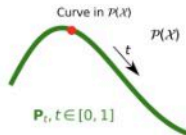
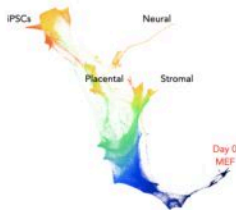


Figure courtesy of Geoffrey Schiebinger

The marginals $\mathbf{P}_t$ describe a curve in $P(\mathcal{X})$, the space of probability measures on $\mathcal{X}$.

How do we **see** the evolution trajectories of the cells?

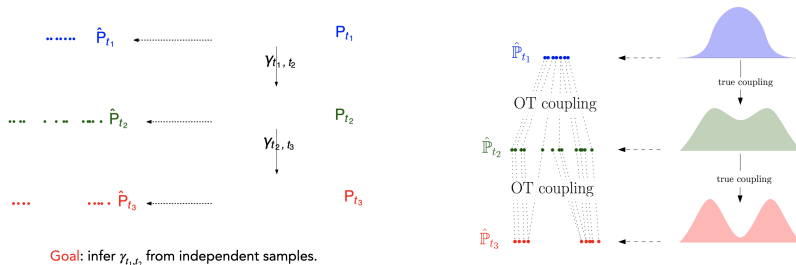
- ▶ **Fundamental Difficulty:** To measure the gene expression, the cell is destroyed.
- ▶ Therefore, we cannot actually see the development of a cell in terms of the evolution of its gene data.
- ▶ So, the developmental process of a cell cannot be measured directly.

- ▶ We only see **snapshots of populations in time**, from the **samples** of different but identical populations.
- ▶ The individual trajectories for cells need to be **inferred** from those snapshots.

- ▶ This is a statistical inference problem:
 - ▶ Given a collection of samples from a distribution, can we guess what the distribution looks like?
 - ▶ Th more samples, the better our guess?

- ▶ **Additional difficulties:**
 - ▶ So many cells. And it is costly to test many cells.
 - ▶ So many genes. The gene expression space has dimension $\geq 20,000$.

- Optimal transport provides a good method for such an inference problem.



Figures courtesy of Geoffrey Schiebinger

- Find optimal solutions $\hat{\pi}_i \in \Pi(\hat{P}_{t_i}, \hat{P}_{t_{i+1}})$.
 - It is about how each cell at time t_i develops to the next gene expression at time t_{i+1} .
- Then, glue together these couplings. $\hat{\pi}_T \circ \hat{\pi}_{T-1} \circ \dots \circ \hat{\pi}_1 \circ \hat{\pi}_0$.

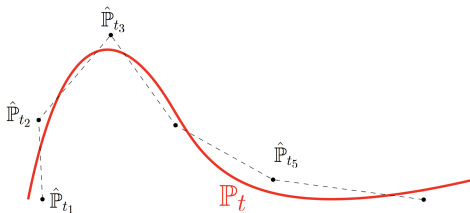
Waddington-OT

When we know only samples $\hat{\mathbf{P}}_{t_1}$ of $\mathbf{P}_{t_i}$, for samples of time $0 = t_0 < t_1 < \dots < t_T = t_{max}$:

Intuition

In the space of distributions with the OT metric,
the process is **locally linear**:

Waddington-OT
[Schiebinger et al.
2019]



It can approximate better and better.

- ▶ By increasing the number of time points T .
- ▶ By increasing number of samples for $\hat{\mathbf{P}}_{t_i}$.

Drawback: If the sample errors are big (e.g. due to small sample size), the solution may suffer overfitting, making it far from the ground truth.

- ▶ An extreme such case is when each sample (at a time point) is a single cell, even if there are many time points.

Optimal transport assumption

Waddington-OT is based on optimal transport assumption of the development.

- ▶ One can directly consider finding a law $\mathbf{P}$,
a probability measure on the space of paths,
that describes the development of each cells over time.
 - ▶ It is a **transport plan** in continuous time:
the particles (i.e. cells) are transported following a certain set of paths over time.

Assumption:

The development follows a law $\mathbf{P}$, a probability measure on the space of paths, corresponding to an SDE of the form

$$dX_t = \nabla \Psi(t, X_t) dt + \sigma dB_t$$

- ▶ This assumption is a short time (dt) and stochastic version (dB_t) of the famous Brenier's theorem, which roughly says

"An optimal transport plan is given by a mapping $T = \nabla \phi$ for a convex function ϕ ."

Question: How do we recover $\mathbf{P}$ from knowing the samples of marginals $\hat{\mathbf{P}}_{t_1}, \dots, \hat{\mathbf{P}}_{t_T}$, especially when each sample size of $\hat{\mathbf{P}}_{t_i}$ is small so subject to significant error?

How to avoid overfitting?

Global Waddington OT (gWOT)

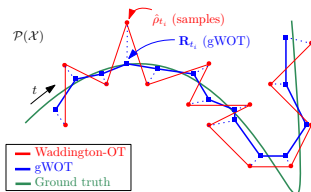
[Lavenant, Zhang, K., & Schiebinger, to appear 2023.]

Let the optimization also find the marginals approximating $\mathbf{P}_{t_i}$'s from the data $\hat{\mathbf{P}}_{t_i}$'s.

- ▶ Combine ‘**regularization**’ and ‘**data fitting**’ together in the functional.
- ▶ **R** = a probability measure on the space of paths
“=” a **transport plan** in continuous time.
- ▶ $\mathbf{R}_{t_i}$ = the marginal distribution of **R** at time t_i “=” the distribution of particles at time t_i that follow the paths governed by the law **R**.
- ▶

$$\min_{\mathbf{R}} \left[\underbrace{\text{Reg}(\mathbf{R})}_{\text{penalize overfitting}} + \frac{1}{\lambda} \underbrace{\text{DataFitting}(\mathbf{R}_{t_1}, \dots, \mathbf{R}_{t_T})}_{\text{penalize deviating from the data}} \right]$$

- ▶ λ = the balance between ‘regularization’ and ‘data fitting’; it is to be guessed.
 - ▶ One may try different values of λ to find the ‘best’ result.



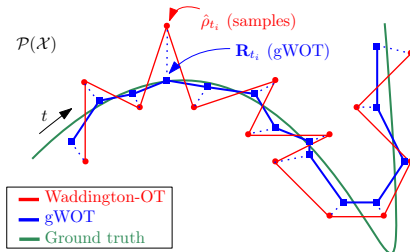
Global Waddington OT (gWOT): consistency

Theorem (Lavenant, Zhang, K., & Schiebinger)

Let $\mathbf{R}^{T,\lambda,h}$ denote the optimal solution of

$$\min_{\mathbf{R}} \left[\underbrace{\text{Reg}(\mathbf{R})}_{\text{penalize overfitting}} + \frac{1}{\lambda} \underbrace{\text{DataFitting}(\mathbf{R}_{t_1}, \dots, \mathbf{R}_{t_T})}_{\text{penalize deviating from the data}} \right].$$

Then,
in the limit $T \rightarrow \infty$, followed by $\lambda \rightarrow 0$, $h \rightarrow 0$,
we have that $\mathbf{R}^{T,\lambda,h}$ converges to $\mathbf{P}$ (narrowly).



- It works even when each sample (at a time point) is a single cell, if there are many (thousands) time points.

Additional information: we skip this

- ▶ The relative entropy: $H(\alpha|\beta) = \int_{\Omega} \log(d\alpha/d\beta) d\alpha$.

- ▶ $H(\alpha|\beta)$ measures how much α deviates from β .



$$Reg(\mathbf{R}) = \sigma^2 H(\mathbf{R}|\mathbf{W}^\sigma)$$

Compare the process $\mathbf{R}$ (a probability measure on the space of paths) with $\mathbf{W}^\sigma$ (the Wiener measure corresponding to the Brownian motion σB_t with diffusivity σ^2 .)



$$\text{DataFitting}(\mathbf{R}_{t_1}, \dots, \mathbf{R}_{t_T}) = \sum_{i=1}^T |t_{i+1} - t_i| \left[H(\hat{\rho}_{t_i}^h | \mathbf{R}_{t_i}) - H(\hat{\rho}_{t_i}^h | \text{vol}) \right]$$

$\hat{\rho}_{t_i}^h$ = a smooth approximation of the empirical data $\hat{P}_{t_i}$ at time t_i .

Further questions:

- ▶ Rate of convergence as we have more and more samples?
- ▶ What is the correct metric (equivalently, cost c), for the space of gene expressions?
 - ▶ In the previous work we used $c(x, y) = |x - y|^2/2$ for $x, y \in \mathbb{R}^d$ for the underlying distance structure on the gene expression space.

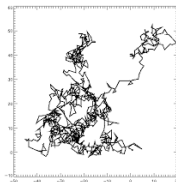
Matching under stochastic constraints

- ▶ Optimal transport reveals certain extremal structures of the matchings.

Matching under stochastic constraints

We may consider matching μ and ν where mass particles are moved in a certain restricted way:

- ▶ Constrain the transport to be a **martingale**:
 - ▶ **Question:** Given two probability measures μ and ν , can we realize ν by a martingale starting from μ , and if so what is the structure? Especially, if we do it in an optimal way?
- ▶ Constrain them to follow **stochastic processes**, e.g. **the Brownian motion**.
 - ▶ **Question:** Given two probability measures μ and ν ,
 - can we realize ν by stopping appropriately the Brownian motion from the distribution μ ,
 - and if so what are the properties of such a stopping time?
Especially, if we do it in an optimal way?



from CRM-physmath

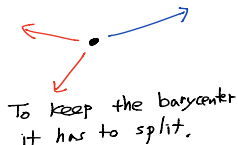
Martingale

Branches out while keeping the barycentre.

(martingale constraint):

$$\int y d\pi_x(y) = x.$$

π is martingale $\implies$ for π to move a unit mass, it has to **split the mass!**



Remark: [Strassen]

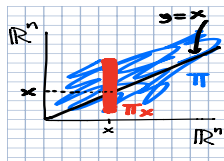
“There is a martingale transport from μ to ν .”

$\Leftrightarrow \mu$ and ν are in convex order;

$\mu \leq_c \nu$, i.e. $\int \xi d\mu \leq \int \xi d\nu, \forall \text{ convex } \xi : \mathbb{R}^n \rightarrow \mathbb{R}.$

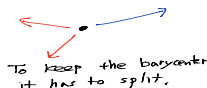
Optimal **Martingale** Transport Problem

- ▶ $MT(\mu, \nu)$ = the set of martingale transport plans from μ to ν



$$\inf_{\pi \in MT(\mu, \nu)} \int_{\mathbb{R}^n \times \mathbb{R}^n} c(x, y) d\pi(x, y).$$

- ▶ For π to move a unit mass, it has to **split the mass**!



Q: What is the **structure** of the set where the **optimal** martingale splits mass?

- ▶ [Hobson & Neuberger] [Begilebock & Juillet] In $\mathbb{R}^1$: Geometric structures of splitting sets, e.g. splitting into two points for $c(x, y) = |x - y|$.
- ▶ [Ghoussoub, K. & Lim] In $\mathbb{R}^n$: Geometric structures of splitting sets; **extremal structures**
 - ▶ e.g. when $\dim = 2$
 - ▶ e.g. in general dimensions if ν can be obtained by a diffusion process from μ .

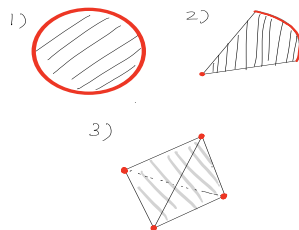
Further questions

Assume:

- ▶ In $\mathbb{R}^n$. $c(x, y) = |x - y|$
- ▶ $\mu \ll \mathcal{L}^n$
- ▶ $\pi \in MT(\mu, \nu)$ be optimal.

Conjecture [Ghousshoub, K. & Lim]

Then, for μ almost every x ,
the splitting set Γ_x
has **extremal** property,
that is, $\bar{\Gamma}_x = \text{Ext} \left(\text{conv}(\bar{\Gamma}_x) \right)$.



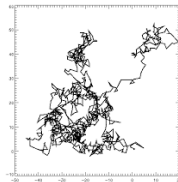
This conjecture is verified

- ▶ e.g. when $\dim = 2$;
- ▶ e.g. in general dimensions if ν can be obtained by a diffusion process from μ ; or
- ▶ e.g. when the target measure ν is discrete.

Optimal transport of stopping Brownian motion

Question: Given two probability measures μ and ν ,

- ▶ can we realize ν by stopping appropriately the Brownian motion from the distribution μ ,
- ▶ and if so what are the properties of such a stopping time?
Especially, if we do it in an optimal way?



from CRM-physmath

- ▶ A **stopping time** τ of Brownian motion is a random time at which one stops a particle following the Brownian motion; at that time, the decision to stop or not is made only using the information of the present and the past, not the future.
- ▶ Notice that stopping time depends on each particular path.
- ▶ For given nonnegative measures μ, ν and a stopping time τ ,

$$"B_0 \sim \mu, B_{\tau} \sim \nu"$$

means that for each measurable set E ,

$$\nu[E] = \int \text{Prob}[B_{\tau} \in E \mid B_0 = x] d\mu(x).$$

Skorokhod problem

For **given** probability measures μ, ν , does there exist a **stopping time** τ of the Brownian motion with $\mathbb{E}[\tau] < \infty$, such that

$$B_0 \sim \mu \quad \& \quad B_\tau \sim \nu?$$

- ▶ [Skorokhod] [Root] [Rost] [Azéma/Yor] [Vallois] [Perkins] [Jacka] ...[Obloj]...

Remark:

- ▶ For such a stopping time τ to exist we need
 - ▶ μ and ν are in **subharmonic** order, $\mu \prec_{SH} \nu$,
i.e. $\int \xi d\mu \leq \int \xi d\nu$,
 $\forall$ subharmonic $\xi : \mathbb{R}^n \rightarrow \mathbb{R}$ ($\Delta \xi \geq 0$).
- ▶ We may view the Brownian motion with a stopping time τ , from μ to ν , as a transport plan.

Connection to optimal transport: **Optimal Skorokhod Problem**

What can we say about an **optimal** stopping time τ for

$$\mathcal{P}(\mu, \nu) := \inf_{\tau} \{ C(\tau) \mid B_0 \sim \mu \text{ \& } B_{\tau} \sim \nu \}?$$

- ▶ (Markovian cost) e.g. $C(\tau) = \mathbb{E} \left[\int_0^{\tau} L(t, B_t) dt \right]$
- ▶ (non-Markovian cost) e.g. $C(\tau) = \mathbb{E} [|B_0 - B_{\tau}|]$,

Literature:

- ▶ [Hobson]
- ▶ [Beiglböck/Cox/Huesmann '13].
 - ▶ **Optimal transport** unifies the previous results on Skorokhod problem, with Markovian cost.
- ▶ [Ghoussoub, K. & Palmer '18–'20]. PDE and dynamic programming methods: dual attainment, general dimensions, Markovian and non-Markovian cost, etc.
- ▶ And many more people.

Optimal Skorokhod problem $\mathcal{P}(\mu, \nu)$

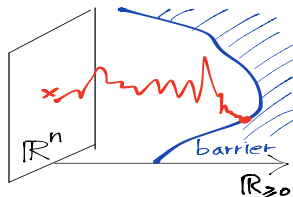
[Beigleböck, Cox, & Huesmann '13]

[Ghoussoub, K. & Palmer '18].

- ▶ geometric structures for the cost $\mathbb{E} \left[\int_0^\tau L(t, B_t) dt \right]$.
- ▶ The optimal stopping time is uniquely determined by hitting a certain **barrier in the space-time** $\mathbb{R}_{\geq 0} \times \mathbb{R}^n$.

- ▶ **Barrier** $R \subset \mathbb{R}_{\geq 0} \times \mathbb{R}^n$
- ▶ The **hitting time** τ^R to R ,

$$\tau^R := \inf \{ t \geq 0 \mid (t, B_t) \in R \}.$$



[Ghoussoub, K. & Palmer '18] determined the optimal barrier R^* by solving the optimal solution to the dual problem (on a certain space of functions).

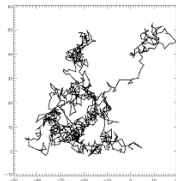
Further questions

- ▶ How can we analyze the properties of the optimal barriers R^* ?
- ▶ The optimal barriers play the role of Brenier's optimal mapping for the stopping time problem.
 - ▶ For Brenier's optimal mapping, we have a corresponding PDE, called the Monge-Ampère equation to study its properties.
 - ▶ For the optimal barrier, what analytical tools can we develop?

Optimal transport approach to free boundary problems

Application of optimal transport methods to free boundary problems

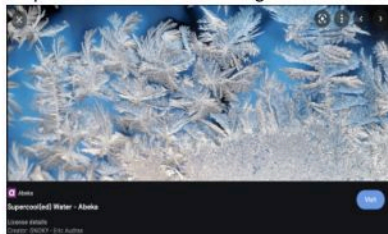
- ▶ An optimal transport approach gives a robust method for studying free boundaries.
- ▶ Optimal transport with the Brownian motion stopping gives the description of the **free boundary** of phase transition phenomena. [Kim & K.]



from CRM-physmath

The Stefan Problem

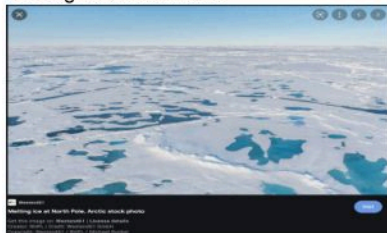
Supercooled water freezing into Ice



source: Google images

unstable behaviour

Melting of ice into water



source: Google images

stable behavior

- ▶ Only few results on the supercooled problem (**the unstable problem**):
 - ▶ 1-D, special case [Chayes/Swindle][Chayes/I.Kim'08][Lacoin '14][Delarue-Nadtochiy-Shkolnikov '19].
 - ▶ self-similar, well-prepared (smooth) radial data [Hadzic/Raphael]
- ▶ Many works for the **stable problem**
 - ▶ Well-posedness: [Meirmanov]
 - ▶ Regularity: [Caffarelli'77], [Athanasopoulos/Cafferli/Salsa '96-98][Choi/I.Kim'10], recently, [Figalli, Cabre, Serra, Ros-Oton]...
 - ▶ Stability: [Hadzic/Shkoller'14][Hadzic/Raphael'16],
- ▶ The supercooled problem (the unstable problem) is poorly understood, while there are many good results for the stable problem.
- ▶ **The supercooled problem (the unstable problem) is not well-posed in general.** There is in general no uniqueness for the solutions, and little is known for its global existence especially in multi-dimensions.

Free target problem (K. & I. Kim)

► [Free target under a density constraint]

Given $f \geq 0$, $f \in L^\infty(\mathbb{R}^n)$ (e.g. $f \equiv 1$).

consider

$$\mathcal{P}_f(\mu) := \inf_{\tau, \nu} \{ \mathcal{C}(\tau) \mid B_0 \sim \mu, B_\tau \sim \nu, \nu \leq f, \nu \text{ is compactly supported} \}$$

Here $\mathcal{C}(\tau) = \mathbb{E} \left[\int_0^\tau L(t, B_t) dt \right]$, the transportation cost along the Brownian motion up to the stopping time τ .

- The optimal solution ν^* **saturates** the upper bound constraint f on the active region, namely,

$$\nu^*(x) = f(x) \text{ "if an active Brownian particle pass by at } x\text{".}$$

- The pair of optimal stopping and the optimal target (τ^*, ν^*) has corresponding moving mass η and stopping rate ρ in the space-time, **solving**

the Stefan problem

$$\partial_t \eta - \frac{1}{2} \Delta \eta = -\rho$$

$$\rho = \pm \partial_t (f \chi_{\{\eta=0\}}) \leftarrow \text{This is due to the saturation result.}$$

‘+’ for the supercooled (unstable) problem, ‘−’ for the usual (stable) problem.

- For the original problem $f \equiv 1$.
- The boundary $\partial(\chi_{\{\eta=0\}})$ is the free boundary at a given time.

- ▶ Our method works for both stable and unstable (supercooled) problems.
- ▶ In particular, we are able to obtain a certain type of well-posedness result for **the supercooled Stefan in general dimensions**.
 - ▶ **Our result is a limited result:**
 - ▶ It solves the supercooled (unstable) problem by **providing a good initial active domain** $E = \lim_{t \rightarrow 0^+} \{x \mid \eta(x, t) > 0\}$ determined by the optimal free target ν^* .

Further questions:

- Our method is robust and can be applied to

Nonlocal Stefan problem:

This is modelled on stochastic processes that allow jump of particles.

As a PDE, it can be written:

(for $0 < s < 1$),

$$\partial_t \eta + (-\Delta)^s \eta = -\rho$$

where

$$\rho := \begin{cases} \partial_t ((1 - \kappa_1) \chi_{\{\eta=0\}}) & \text{(supercooled) ,} \\ \partial_t ((-1 + \kappa_2) \chi_{\{\eta=0\}}) & \text{(the usual).} \end{cases}$$

- Here, κ_1, κ_2 arise due to **nonlocal effect**: the stochastic particles that carries heat can jump across the free boundary and stop.
- Recent result [Chu, K., Kim, & Nam].
- **A difficult open problem** is to solve the supercooled Stefan problem **where the initial active domain** $E = \lim_{t \rightarrow 0^+} \{x \mid \eta(x, t) > 0\}$ **is prescribed**.
 - A result to be announced soon. [Choi, K. & Kim].
- Regularity of free boundary for a certain class of solutions?
 - This is related to developing methods for analyzing the optimal barriers in the optimal stopping time problems.



<https://kantorovich.org>

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- ▶ This is an interdisciplinary initiative started by people in Pacific Northwest region in North America (Vancouver, Seattle, Edmonton, Saskatoon), and in various areas (mathematics, statistics, economics, computer science, biology) **connected by optimal transport theory**.
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Thank you!